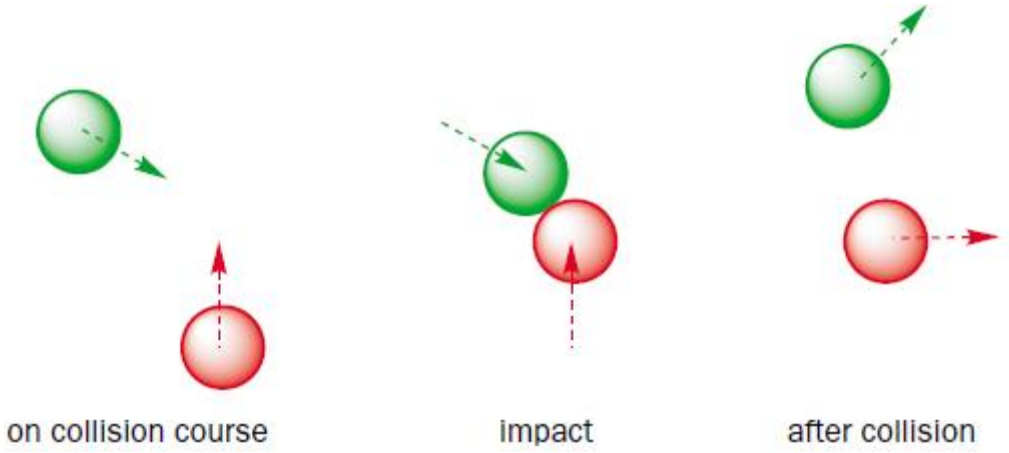
ORGANIC REACTIONS

- **Why molecules generally don't react with each other!**
- **Why sometimes molecules do react with each other?**
- **In chemical reactions electrons move from full to empty orbitals**
- **Molecular shape and structure determine reactivity**
- **Representing the movement of electrons in reactions by curly arrows**

Organic Reactions: Breaking bonds and forming new bonds

- **Electron flow is the key to Reactivity**
- **Molecular Orbital Energy and Shape**
- **Using Curly Arrows Correctly**
- **Guide to Drawing Mechanisms**

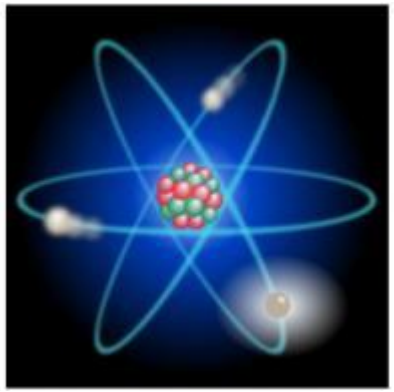
Not all collisions of molecules lead to chemical change!





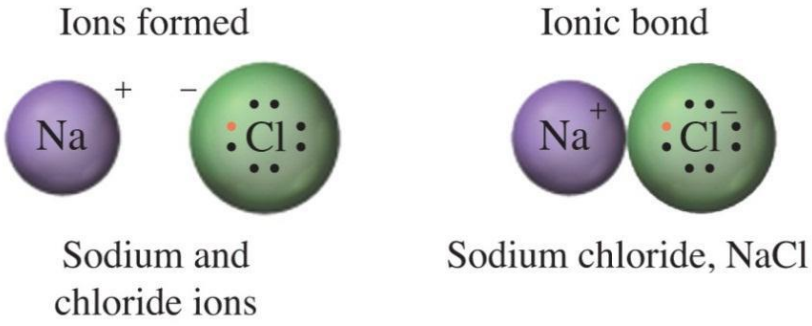
What leads to the possibility of a chemical reaction?

CHARGE



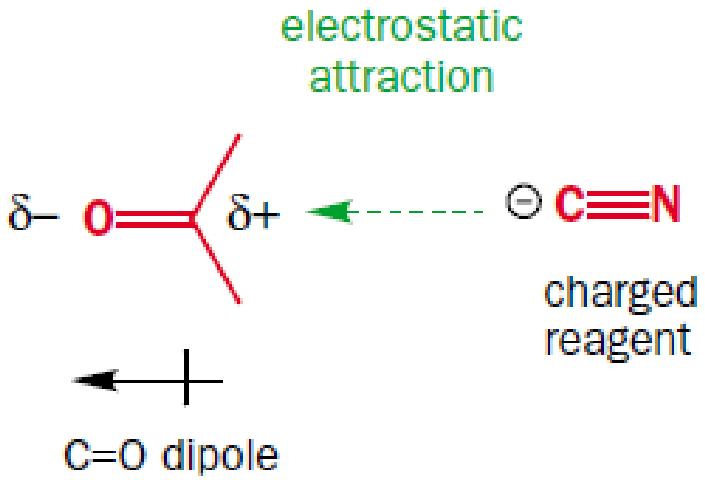
Organic molecules repel each other because of the outer layer of electrons

Inorganic style of electrostatic attraction is rare in organic chemistry



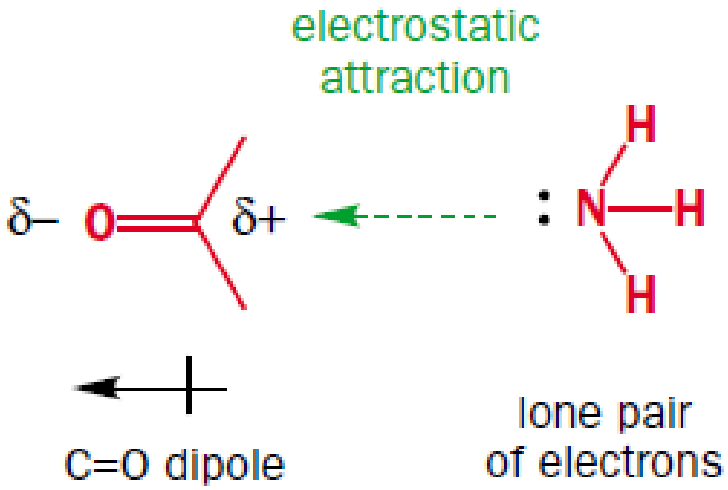
What leads to the possibility of an organic reaction?

PARTIAL CHARGE



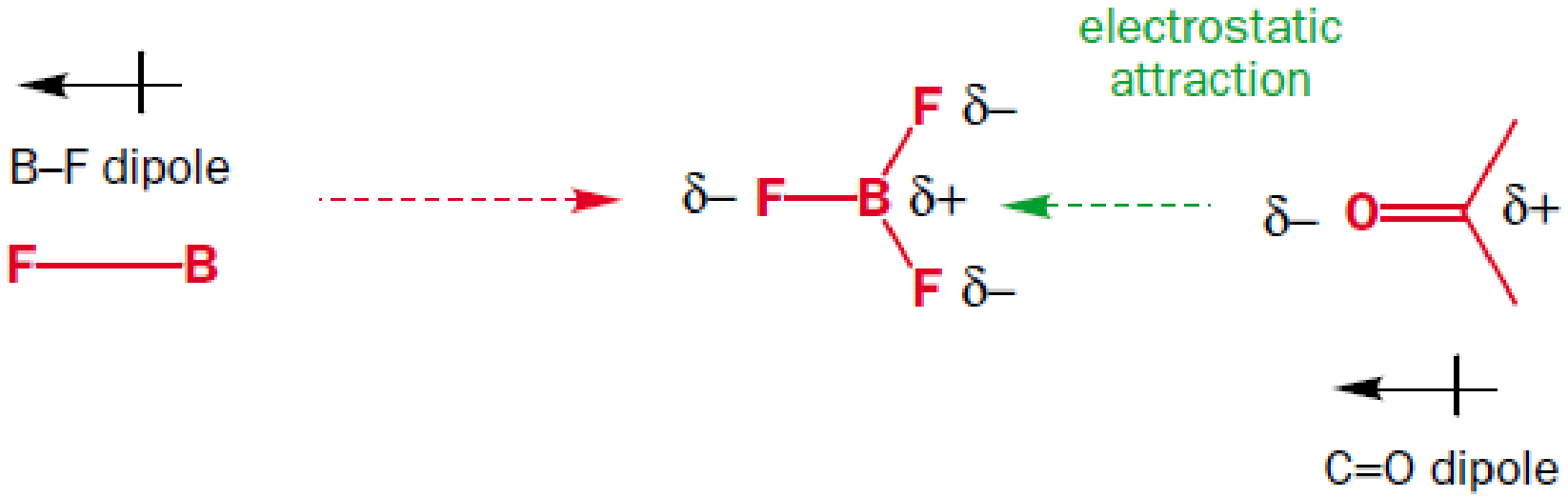
Attraction between a charged reagent-here an anion-and an organic compound that has a dipole moment because of different electronegativities

Interaction can also take place between the polarized C=O bond and an electron-rich species such as NH_3



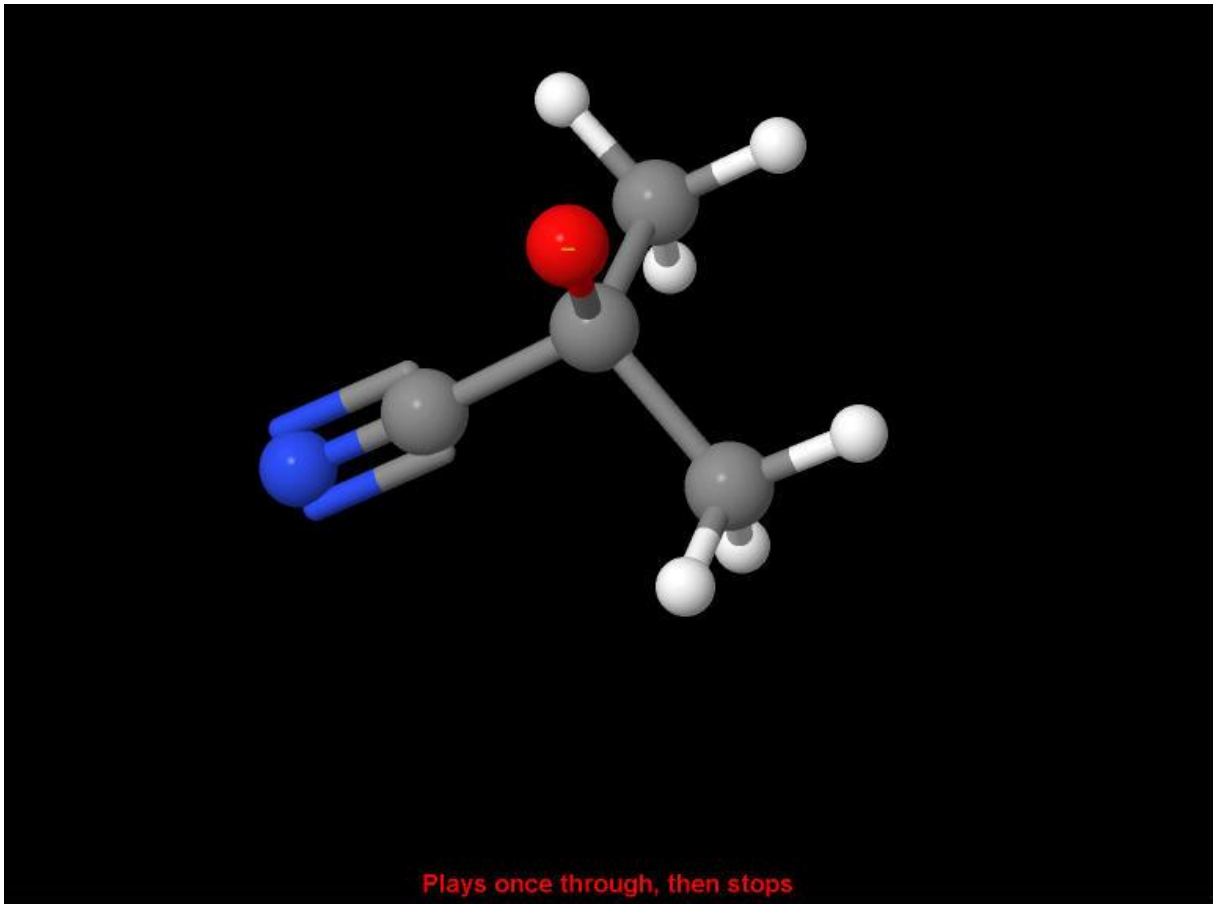
What leads to the possibility of an organic reaction?

PARTIAL CHARGE



Interaction can also take place between the Oxygen-end of the polarized C=O bond and an electron-deficient species such as BF_3. Electrons can move from Oxygen to Boron empty p orbital.

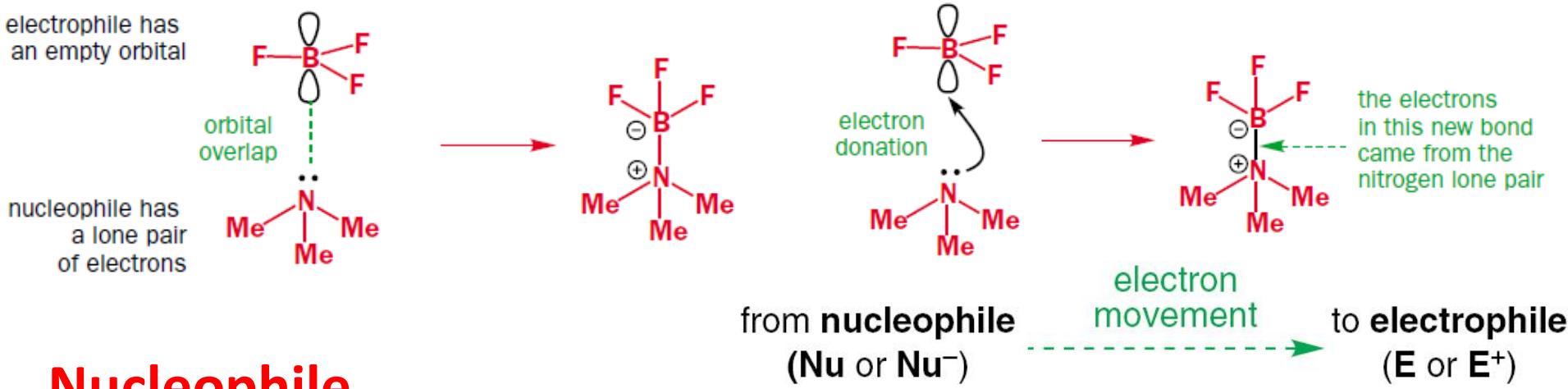
Cyanohydrin Formation



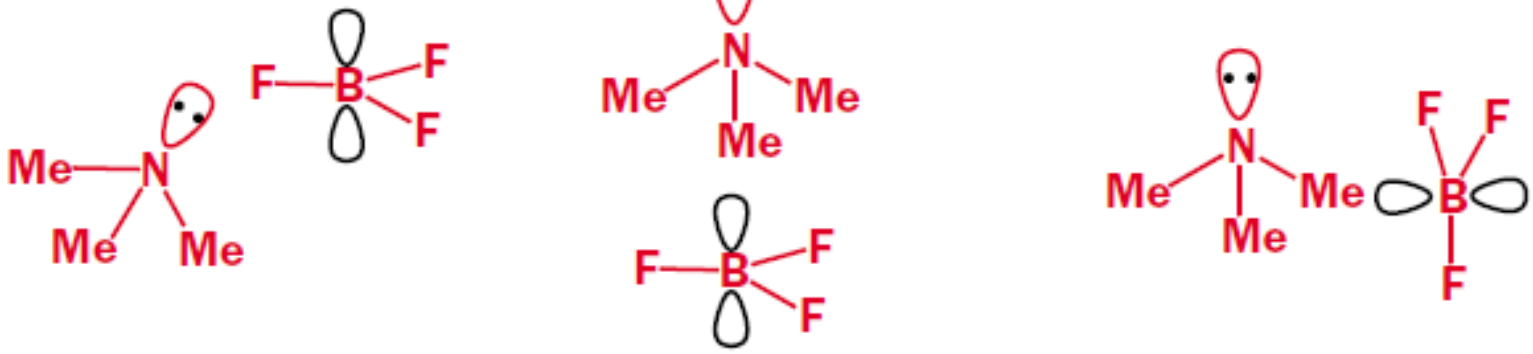
http://www.chemtube3d.com/playground/playground.html
http://www.chemtube3d.com/Nucleophilic%20substitution%20at%20the%20carbonyl%20group%20-%20Cyanohydrin%20formation.html

Where do electrons go?

Electrophile

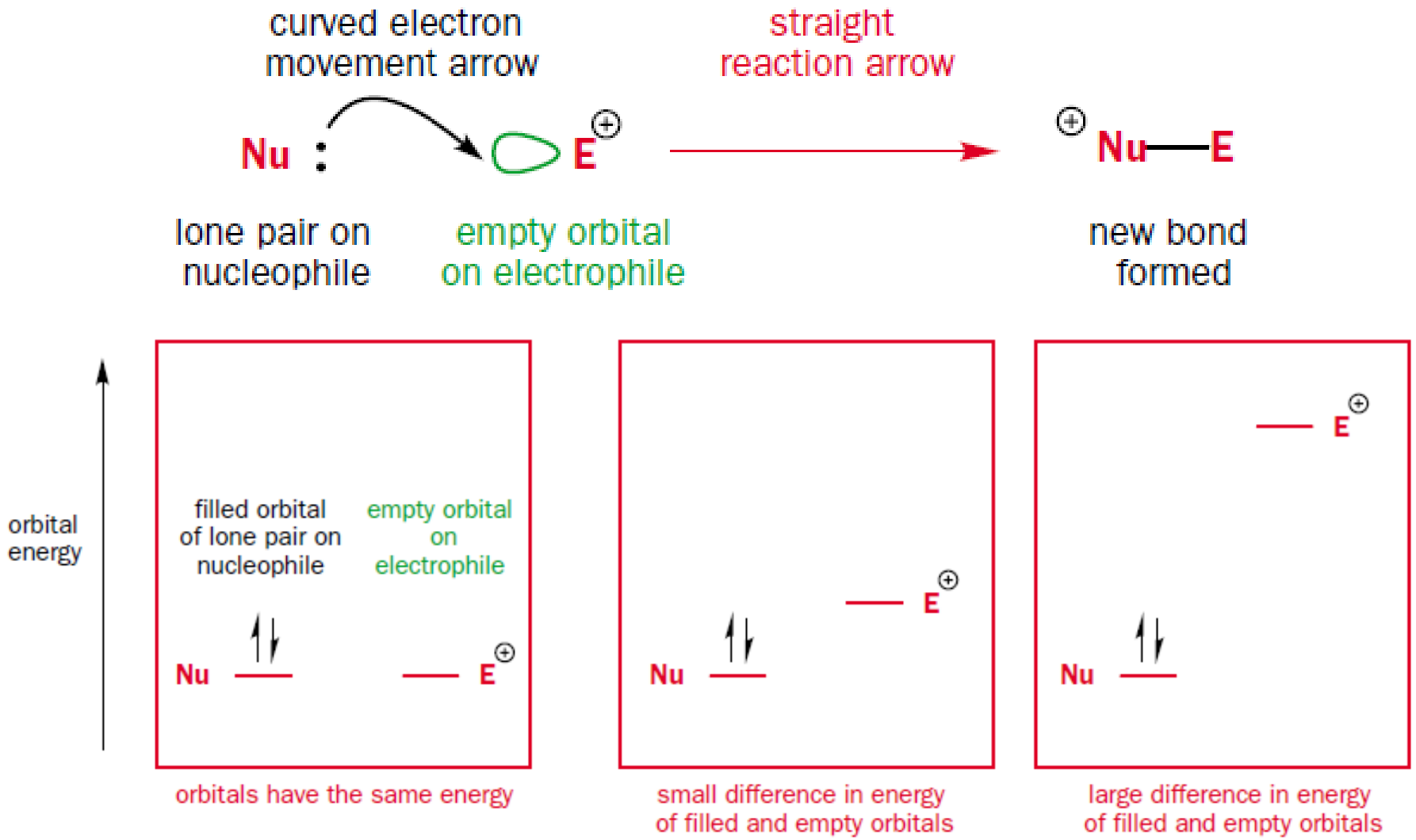


Nucleophile

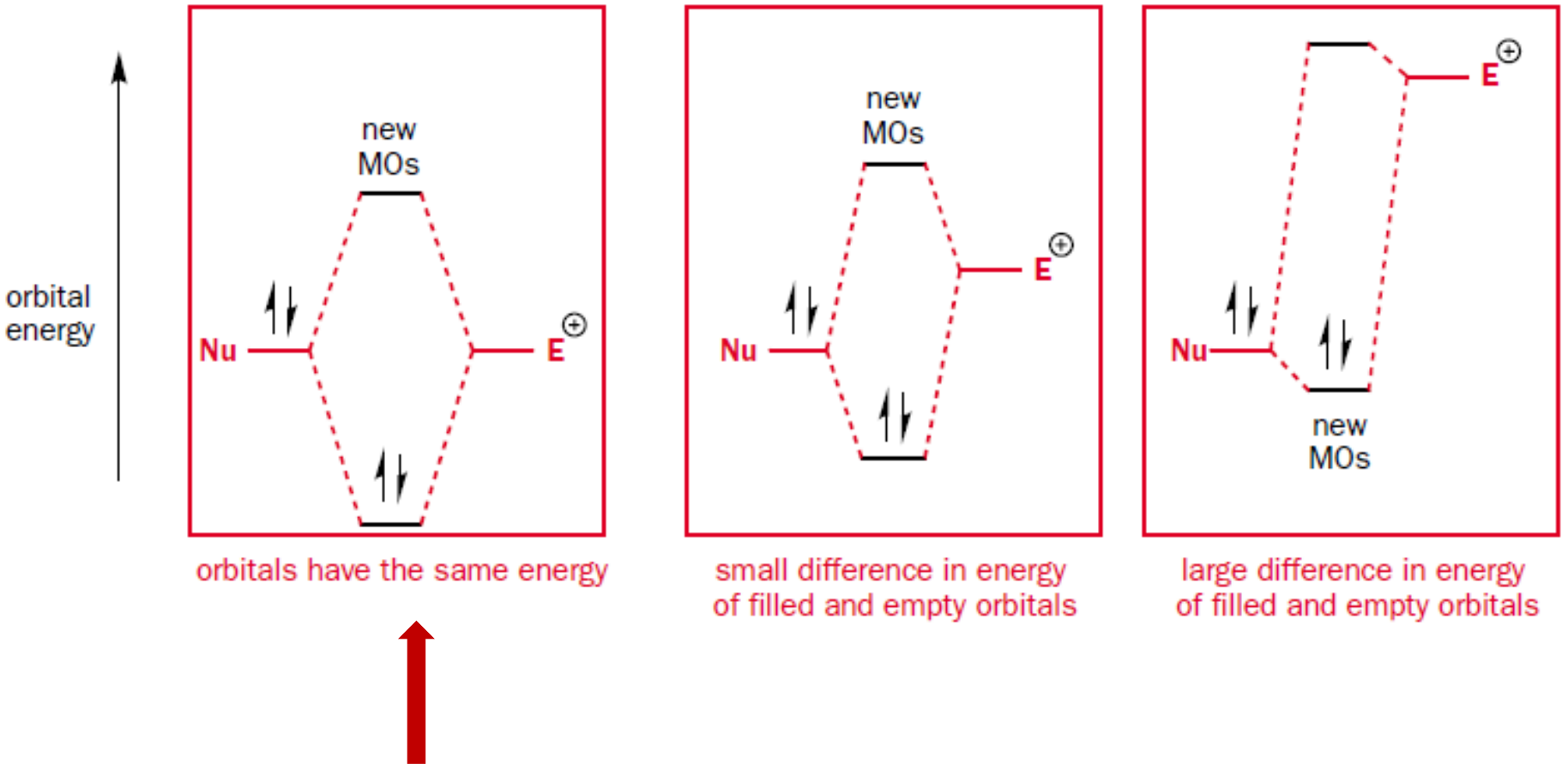


Unproductive arrangements: Unsuccessful orbital overlapping

Relative Energy of Molecular Orbitals is Important



Relative Energy of Molecular Orbitals is Important



Favorable Arrangement:
Largest Energy drop

Organic Reactions-Summarizing…

Molecules repel each other because of their outer coatings of electrons.

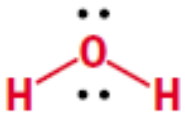
Molecules attract each other because of:

- **attraction of opposite charges**
- **overlap of high energy filled orbital (HOMO) with low energy empty orbitals (LUMO)**

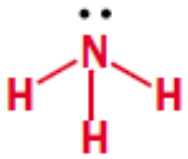
For reaction, molecules must approach each other so that they have:

- **enough energy to overcome the repulsion**
- **the right orientation to use any attraction**

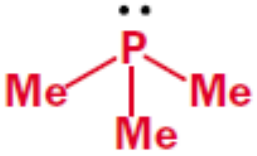
Nucleophiles (Nu)-Examples



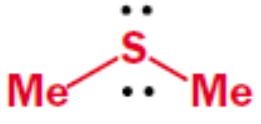
water



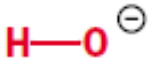
ammonia

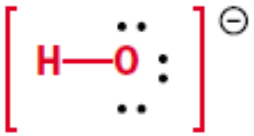


trimethylphosphine



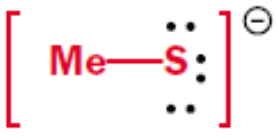
dimethylsulfide





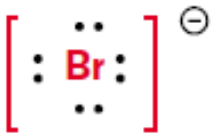
hydroxide



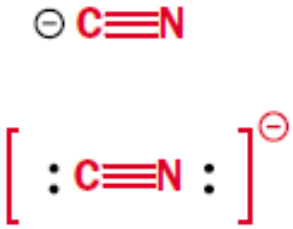


methane thiolate

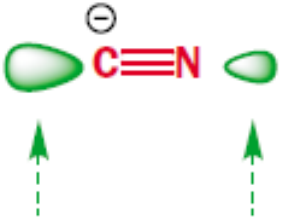


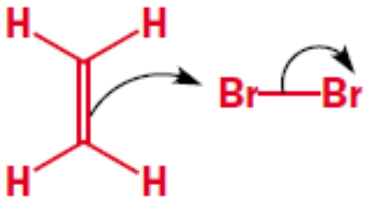


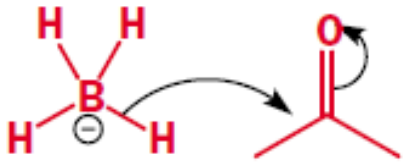
bromide



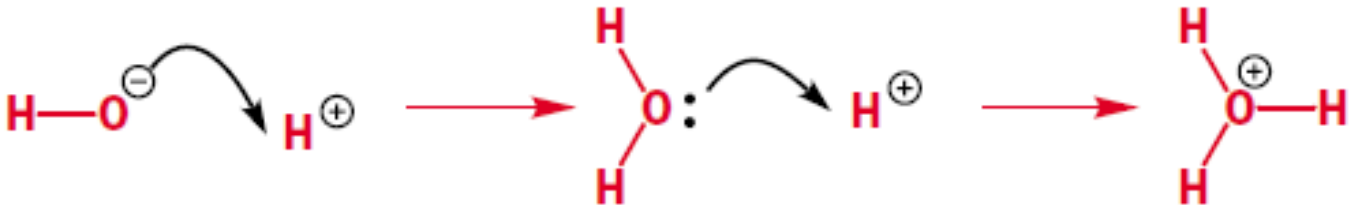
cyanide ion

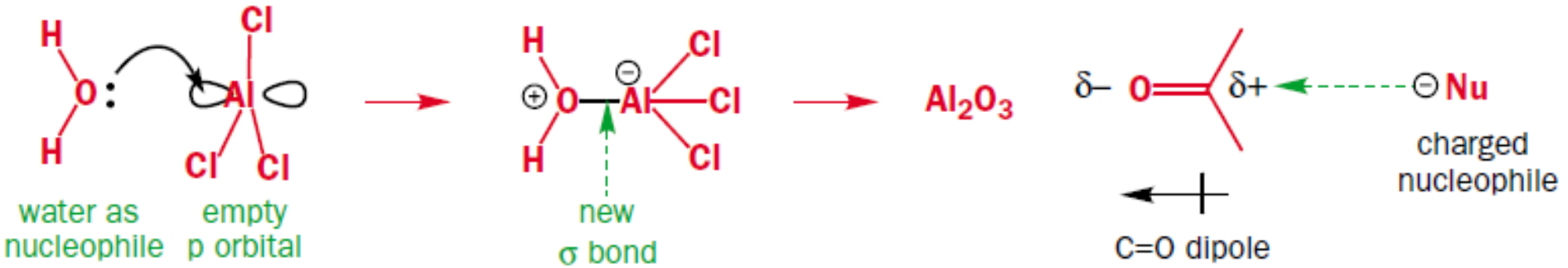






Electrophiles (E)-Examples





What do curly arrows mean?

Curly arrows show the MOVEMENT of a pair of electrons

- **The tail of the arrow shows the SOURCE of the electron pair of the NUCLEOPHILES:**

A filled orbital (HOMO) such as a lone pair or a negative charge or a π bond or a σ bond to an electropositive atom.

- **The head of the arrow shows the DESTINATION of the electron pair to the ELECTROPHILES:**

An empty orbital (LUMO) such as a positive charge or a neutral molecule with an empty p orbital or a π bond or a σ bond to an electronegative atom.

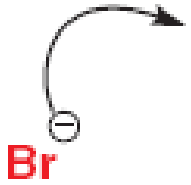
- **Charge is conserved in a reaction.**
- **You must break a bond in the same step when you make a new bond with an uncharged H, C, N or O.**

Nucleophiles and Electrophiles

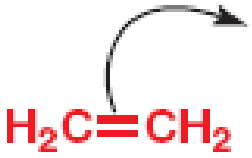
Nucleophiles: Source of electrons



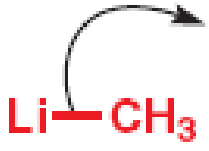
a lone pair



a negative charge



a double bond

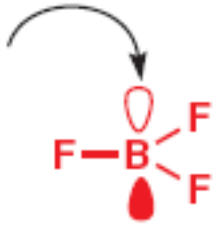


a σ bond to an electropositive atom

Electrophiles: Destination of electrons



a positive charge representing an empty orbital



a neutral molecule with an empty p orbital

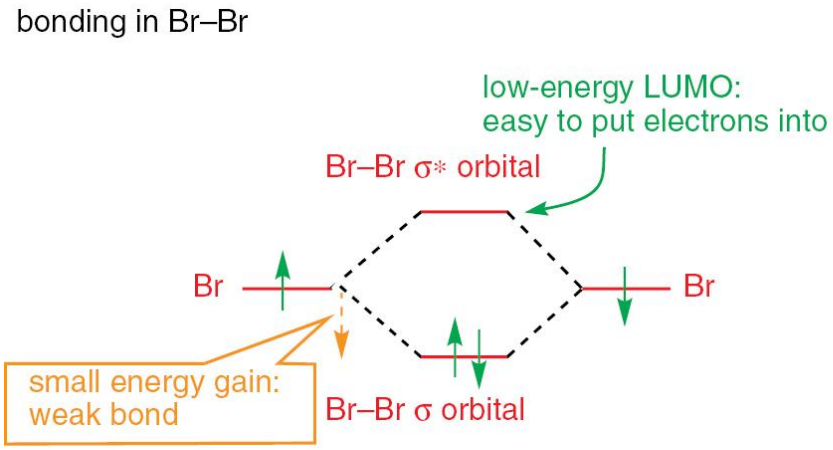


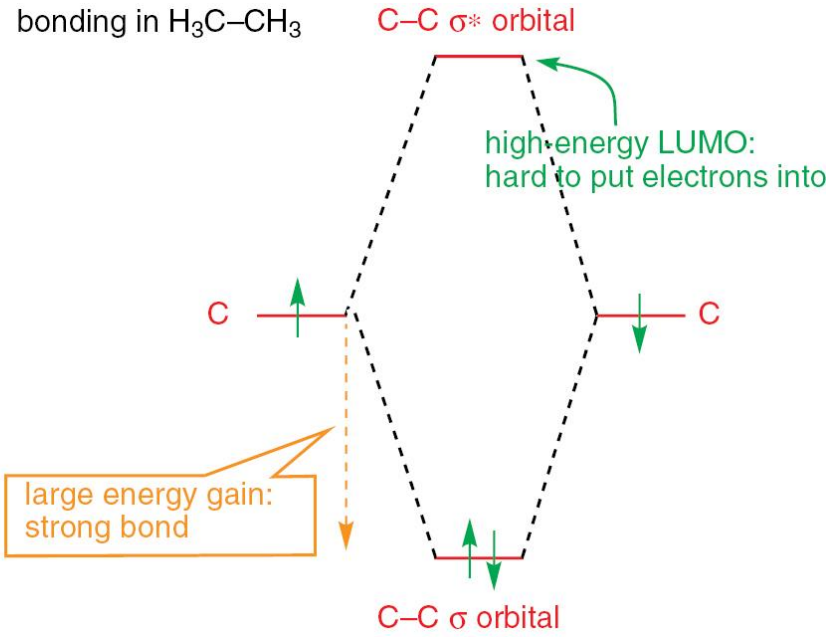
a double bond to an electronegative element



a single bond to an electronegative element

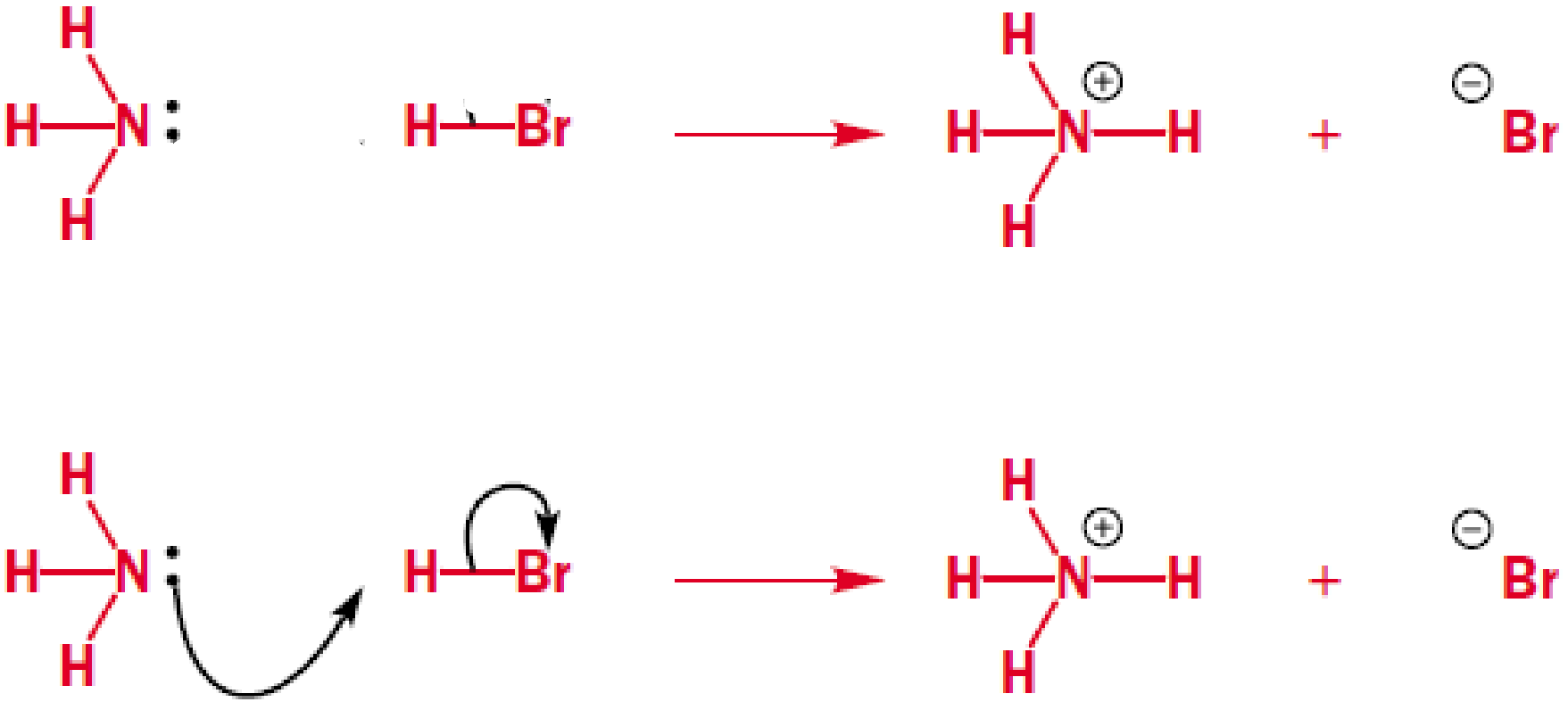
Electrophilic sigma bonds



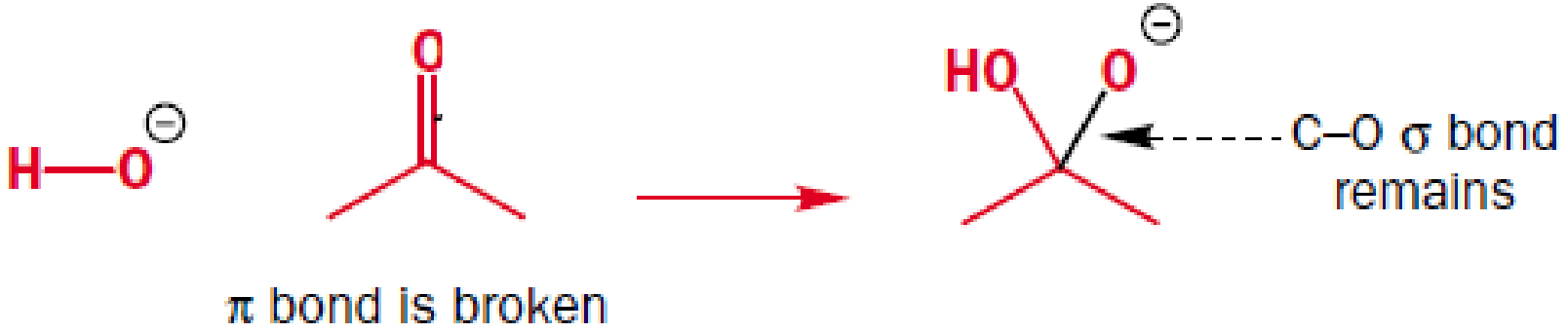


Bromine reacts with Nucleophiles, but alkanes don't!

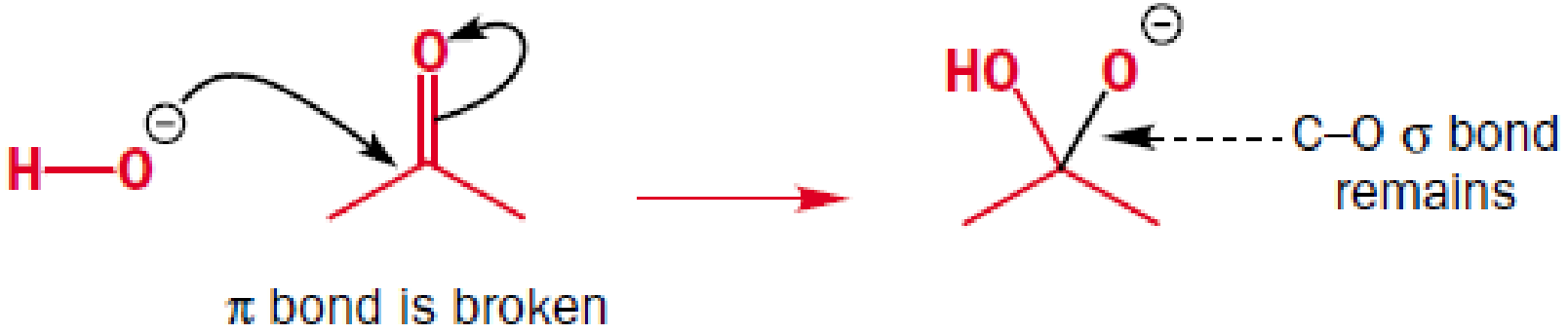
Practice!



Practice!



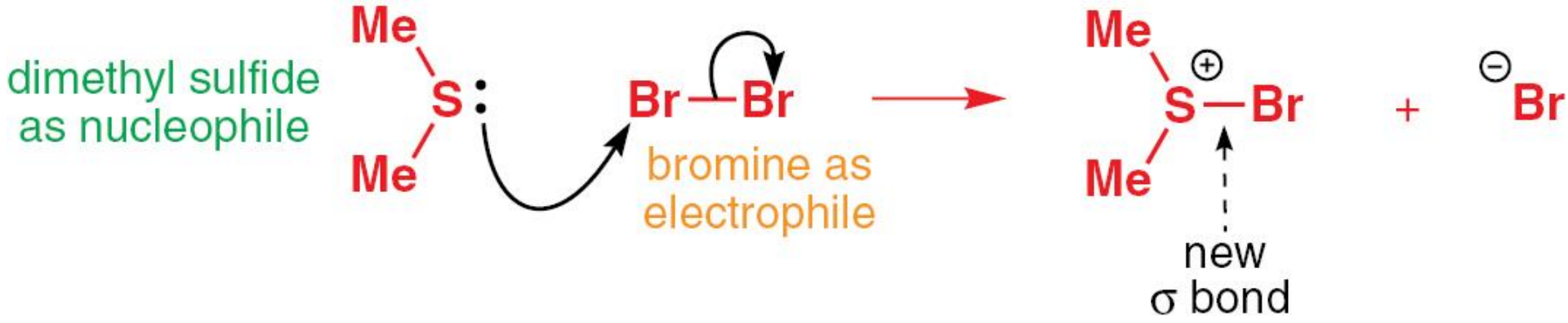
π bond is broken



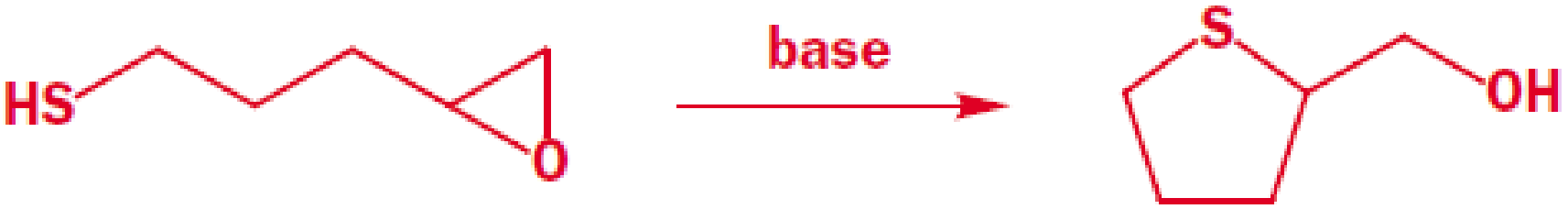
π bond is broken

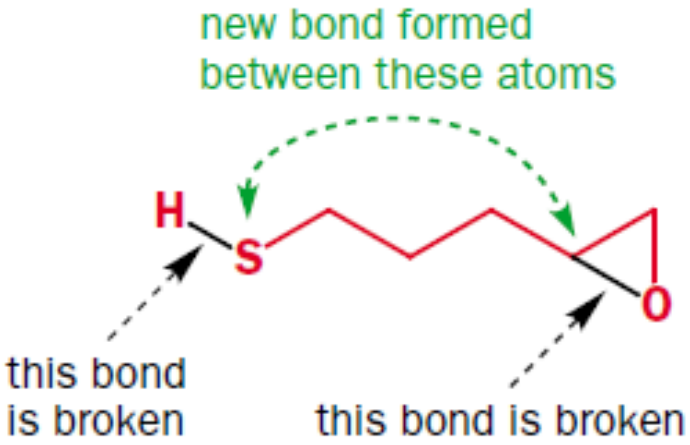
Practice!


dimethyl sulfide
as nucleophile
Me
S
Me
Br—Br
bromine as
electrophile
Me
S—Br
Me
new
σ bond
Br

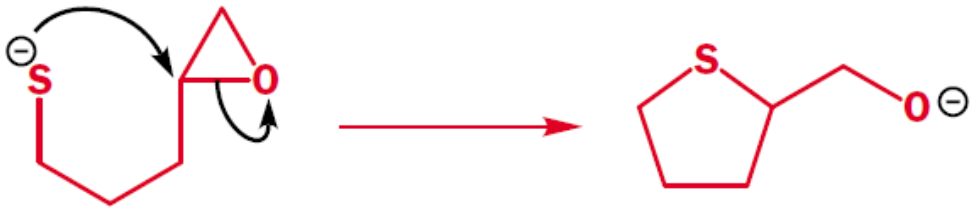

dimethyl sulfide
as nucleophile
Me
S
Me
Br—Br
bromine as
electrophile
Me
S—Br
Me
new
σ bond
Br

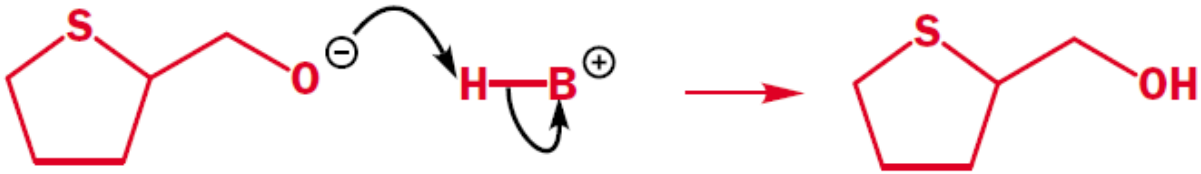
Practice!


HS
O
base
S
OH


new bond formed
between these atoms
H
S
O
this bond
is broken
this bond is broken


HS
O
base
S
O


S
O
S
O


S
O
H
B
S
OH

How to write right mechanisms? Practice!

1. Draw the reagents as clear structures considering the conditions given (acid, base, solvents).
2. Inspect the starting materials and the products. Consider new bonds, broken bonds, moved bonds, new groups added, old groups removed.
3. Find the nucleophilic and electrophilic centres and decide which is the most reactive.
4. If their combination leads to the products, draw the reactants, complete with charges, position them within bonding distance considering the orbitals involved.
5. Draw a curly arrow from the nucleophile, by starting from the source of e's, that is the bond or charge or lone pair, towards the electrophile's empty orbital, that will be the position of the arrow's head.
6. Consider if any changed atom has too many bonds, if so then break one. Draw another curly arrow from the centre of the chosen bond, the filled orbital, and terminate it in a suitable place, such as an electronegative atom.
7. Write out the structures of the products specified by the curly arrows by breaking the bonds that are the sources of the arrows and making those that are the targets. Check the charges. If the structure is wrong, then the curly arrows were wrong so go back and change them.
8. Repeat stages 5 - 7 so as to produce a stable product.

Homework 3



Study Chapter 5 in textbook!